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Source: Rangeland Ecology and Management, 72(6) : 996-1004

Published By: Society for Range Management

URL: <https://doi.org/10.1016/j.rama.2019.07.003>

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Plant Species Composition and Forage Production 14 Yr After Biosolids Application and Grazing Exclusion

E. Avery^a, M. Krzic^{a, b, *}, B.M. Wallace^c, R.F. Newman^c, G.E. Bradfield^d, S.M. Smukler^a

^a Faculty of Land and Food Systems, University of British Columbia, Vancouver, BC V6T 1Z4, Canada

^b Faculty of Forestry, University of British Columbia, Vancouver, BC V6T 1Z4, Canada

^c BC Ministry of Forests, Lands and Natural Resource Operations & Rural Development, Kamloops, BC V2C 2T3, Canada

^d Department of Botany, University of British Columbia, Vancouver, BC V6T 1Z4, Canada

ARTICLE INFO

Article history:

Received 31 December 2018

Received in revised form 5 July 2019

Accepted 11 July 2019

Key Words:

bluebunch wheatgrass
Kentucky bluegrass
native grassland
restoration
soil carbon

ABSTRACT

This paper examines the effects of a single surface application of biosolids (at 20 dry Mg ha⁻¹) on plant species composition, forage quality and quantity, and C stocks after 14 yr of rest in rangelands of the Central Interior of British Columbia. More than two times the aboveground biomass of grasses and the percent cover of plant litter were found in the biosolids treatment relative to the control, along with reductions in bare soil and microbiotic crust cover. Significantly greater plant uptake of all macronutrients (C, N, P, K, S, Ca, and Mg); most micronutrients (B, Cu, Mn, Mo, and Zn); and Al occurred in the biosolids treatment. P and Cu were the only two nutrients to be more concentrated in the biosolids-treated forage relative to the control forage, while N, Mg, and protein were more concentrated in the control forage. No significant difference in forage digestibility was found between biosolids and control treatments. Bluebunch wheatgrass, the late-seral native grass species, had significantly increased cover and aboveground biomass in the biosolids treatment relative to the control; however, between 2006 and 2016, non-native Kentucky bluegrass had reached > 25% cover in the biosolids plots, perhaps restricting the full recovery of bluebunch wheatgrass. Our findings indicate that biosolids application to ungrazed rangeland can increase long-term forage production and reduce bare soil. However, at our study site biosolids application also led to a long-term shift in the plant community composition away from the late-seral (i.e., bluebunch wheatgrass) trajectory, and the effects of this shift on rangeland health and productivity require further investigation.

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INTRODUCTION

Many grasslands of North America have been substantially altered following the introduction of livestock and other agricultural practices by European settlers in the 19th century and no longer support the forage levels that they once did (Suttie et al., 2005; Wang et al., 2014; Chen et al., 2017; Svejcar et al., 2017). Once degraded, semiarid grasslands are slow to recover in terms of both aboveground net primary productivity (ANPP) and species composition (Wikeem and Wikeem, 2004), even when long periods of rest have been enforced (Smoliak et al., 1972). Organic amendments such as biosolids (solid by-product of municipal wastewater treatment facilities) are sometimes applied as fertilizer to rangelands because of their high levels of organic matter and nutrients

(e.g., N and P) that offer valuable resources for these nutrient- and water-limited ecosystems (Pierce et al., 1998; Ippolito et al., 2010). Although increased ANPP following nutrient amendments is a benefit to cattle (De Schrijver et al., 2011), a shift in species composition or reductions in species richness (Newman et al., 2014; Seabloom et al., 2015; Blumenthal et al., 2017; DeMalach, 2018; Melts et al., 2018) may not be the desired outcome for all restoration efforts.

Higher forage yields due to improved supply of nutrients and water holding capacity have consistently been documented during the 3–5 yr after biosolids application (Jurado and Wester, 2001; Martínez et al., 2003; McDougall et al., 2008; Jurado-Guerra et al., 2013) and even up to 12 and 14 yr after biosolids application (Sullivan et al., 2006; Ippolito et al., 2010; Wester et al., 2011). Other studies (Blumenthal et al., 2017) have found that effects of organic amendments on forage productivity declined in the longer term. Hence, it remains unclear how long the effect on forage production can last. Similarly, while biosolids generally improve forage quality and protein content in the short term (McFarland et al., 2007), the

* Correspondence: Maja Krzic, Faculty of Land and Food Systems and Forestry, University of British Columbia, Vancouver, BC V6T 1Z4, Canada.

E-mail address: maja.krzic@ubc.ca (M. Krzic).

long-term relationship between biosolids and forage quality is not well documented.

Plant species composition is an important factor to consider in terms of forage quality, as well as ecological integrity and carbon storage. For example, Wallace et al. (2016) found increased plant production 8 yr after a single biosolids application, but the increases came from annual and perennial forbs at the expense of perennial grasses, lowering the quality of the forage. A recent study on the long-term effects of organic matter applications to degraded rangeland (Blumenthal et al., 2017) has demonstrated that deleterious shifts in plant community composition can occur more than 22 yr after the application, reinforcing the importance of long-term studies that consider plant species composition. The expansion and dominance of Kentucky bluegrass (*Poa pratensis* L.) into native grasslands of the Northern Great Plains (e.g., Printz and Hendrickson, 2015; Sanderson et al., 2016) is an example of how a recent shift in species dominance can have a negative effect on water and nutrient cycling, as well as wildlife habitat (Toledo et al., 2014). The effects of Kentucky bluegrass and its thatch (i.e., continuous mat of roots and litter) on soil C storage have not received much attention and require future investigation.

The objective of this study was to assess the effects of a single surface application of biosolids on plant species composition, forage quality and quantity, and C stocks after 14 yr of rest from grazing in a degraded grassland of the Central Interior of British Columbia (BC). The reason for examining these effects was to determine if land application of biosolids is a beneficial practice for the longer-term restoration of these semiarid ecosystems.

MATERIALS AND METHODS

Site Description

The study was conducted in 2016 on a long-term field experiment established in 2002 at a working ranch near Jesmond, in the South Cariboo region of BC (51°25'N, 122°09'W; elevation 1 100 m above sea level). The soil was an Orthic Dark Brown Chernozem (Typic Haploboroll) formed on glacial moraine overlaid by a thick layer of aeolian deposits (Wikeem and Wikeem, 2004), with a loam texture (14% clay, 46% silt, and 40% sand) and negligible proportion of coarse fragments and carbonates within the main rooting depth (0–30 cm).

The study site was in the grassland phase of the Very Dry Mild Interior Douglas-fir (IDFxm) biogeoclimatic subzone (Wikeem and Wikeem, 2004). The dominant late-seral vegetation of this zone is bluebunch wheatgrass (*Pseudoroegneria spicata* [Pursh] A. Löve), spreading needlegrass (*Achnatherum richardsonii* [Link] Barkw.), and short-awned porcupinegrass (*Hesperostipa curisetia* [Hitchc.] Barkworth), along with an understory of native forbs and cryptogams. When the field experiment was established in 2002, 26% of the site area had exposed mineral soil and the vegetation was dominated by needle-and-thread grass (*Hesperostipa comata* [Trin. & Rupr.] Barkworth), with secondary species of Junegrass (*Koeleria macrantha* [Ledeb.] J.A. Schult. f.), low pussytoes (*Antennaria dimorpha* [Nutt.] T. & G.), white pussytoes (*Antennaria microphylla* Rydb.), and prairie sagewort (*Artemisia frigida* Willd.), collectively indicating an early to midseral stage due to historical moderate-to-heavy grazing practices (Newman et al., 2014).

This region is located in the rain shadow of the Coast Mountains, receiving a total of 415 mm precipitation annually, of which 146 mm falls as snow. The highest precipitation rates occur in late spring or early summer (May, June, and July). In the summer, dry heated air rises from the Fraser river valley and contributes to dry conditions on the adjacent plateau, which are interrupted by intense precipitation events (Wikeem and Wikeem, 2004). The mean annual temperature is 3.8°C, while the mean January and July temperatures are −7.8°C and 14.8°C, respectively. The region receives 101 frost-free d per yr

and 1 174 growing degree days (base 5°C) (Climate BC, 2017). In the yr of data collection for this study (2016), the site received 276 mm of precipitation from May to September, which was 100 mm more than the long-term average (30 yr) measured at a nearby (28 km) weather station (Wildfire Service, 2018).

In the fall of 2002, establishment of the long-term field experiment began with the selection of four randomly located blocks with uniform plant species composition and cover (Newman et al., 2014). Each block (60 m × 70 m in size) was fenced with a 1.5 m-high barbed wire to prevent cattle grazing. The experiment was laid out in a randomized complete block design with two treatments: 1) no biosolids (control) and 2) single surface application of biosolids at 20 dry Mg ha^{−1} replicated once in each of the four blocks. The biosolids were surface applied using a pull-type manure spreader (Hydra-Spread) without subsequent incorporation. The biosolids were supplied by the Annacis Island wastewater treatment plant in Metro Vancouver, BC where the Class A biosolids had been anaerobically digested and dewatered. They contained total N and P contents of 54.9 g kg^{−1} and 30 g kg^{−1} (dry weight), respectively. More detailed information on the biosolids used in this study can be found in Wallace et al. (2009).

Sampling and Analysis

Plant Sampling

Plant species composition was assessed on 20–21 June, 2016, using canopy cover sampling (Daubenmire, 1959) and following the same methodology and locations that were used throughout the field experiment (2002–2016). Percent canopy cover of individual vascular plant species and four ground cover types (mineral soil, microbiotic crust, litter, and biosolids) was estimated within 10 Daubenmire frames (0.2 m × 0.5 m) located randomly along five 50-m-long transects for a total of 50 subsamples per experimental unit. Transects were spaced 5 m apart to provide balanced coverage within each experimental unit. Plant species composition was initially determined in June/July 2002 (i.e., before biosolids were applied) and annually in the same months thereafter for 4 yr (2003–2006; Newman et al., 2014) until the latest measurement in 2016, which was the focus of this study. Plant species nomenclature followed Douglas et al. (1998–2001). Detailed measurements of bluebunch wheatgrass vigor was assessed using one randomly selected bluebunch wheatgrass individual along each of the five 50-m-long transects for a total of five individuals per experimental unit. The circumference and height were measured, and the number of flower stalks on each individual was counted. The green standing biomass of each individual was then clipped at the soil surface, oven-dried (60°C for 24 hr), and weighed to determine standing biomass (g per plant).

Total green standing biomass was clipped at the soil surface within a 1-m² frame on 20–21 June 2016 and separated into two functional groups of grasses and forbs. Five frames were clipped per experimental unit. The plant material was oven-dried (60°C for 24 hr) and weighed by functional group to assess aboveground biomass (kg ha^{−1}). Tissue nutrient analysis was done on the total aboveground biomass from each frame (grasses and forbs combined), ground together, and passed through a 2-mm mesh screen (Wiley Mill. N, C, and S were determined by dry combustion (Nelson and Sommers, 1996) using LECO analyzer (LECO Corp., St. Joseph, MI), while the remaining macronutrients and micronutrients (B, Ca, Cu, Fe, K, Mg, Mn, Mo, Na, P, and Zn) and Al were determined by microwave digestion followed by inductively coupled plasma optical emission spectrometry (Kalra and Maynard, 1991). Plant tissue macronutrients and micronutrients are presented as both tissue concentrations (mg kg^{−1}) and total nutrient uptake (kg ha^{−1}), which was calculated as the product of nutrient concentration in plant tissue and dry matter yield (Jarrell and Beverly, 1981). Total aboveground biomass (as above) was also analyzed for acid

detergent fiber (ADF) to determine amounts of fiber, lignin (ash free), and cellulose in the forage (Goering and Van Soest, 1970). Protein content was analyzed by acid hydrolysis followed by colorimetric analysis (Marks et al., 1985).

Thatch Sampling

Thatch is composed of a layer of dead and living stems (tillers, stolons, and rhizomes), roots, and enmeshed soil (Smiley, 1981). Thatch sampling was performed along the five plant transects per experimental unit, at five random locations per transect. At each location, depth (cm) of thatch was recorded, or 0 was recorded if thatch was not present. At each location that thatch was present, a 10 × 10 cm area of thatch was sliced with a flat shovel and a thatch sample, which included the entire depth of the thatch, was carefully extracted. In the laboratory, thatch samples were cut to a known volume (using a sharp-edged cylinder), weighed, dried in the oven at 60°C for 24 hr, and weighed again to determine oven-dry thatch density. Because all thatch originated from Kentucky bluegrass, total thatch biomass (kg ha^{-1}) was calculated as the product of Kentucky bluegrass cover, average depth of thatch, and thatch density.

To prepare thatch samples for C analysis, plant and soil components were physically separated using a 2-mm sieve. Soil and roots were weighed separately to determine the portion (by mass) of the thatch that each represented. Roots were washed with deionized water, and C concentration of soil and roots was determined by dry combustion (Nelson and Sommers, 1996). Total soil C and N were also determined using the dry combustion method, and contents of the total soil C and N were reported in Avery et al. (2018).

Statistical Analysis

All data were analyzed as a randomized, complete block design with four replications. Subsamples collected within treatment plots were averaged by treatment plot before analysis. All statistical analysis was performed using R statistical software version 3.4.0 (R Core Team, 2017). Mixed-effect models were constructed using the *nlme* package (Pinheiro et al., 2017). For data that were sampled only once (forage quality, aboveground biomass, and aboveground and thatch C stocks), treatment (biosolids or no biosolids) was the only fixed effect in the model. For the plant species composition and soil cover data, which were assessed over multiple yr (2002–2006 and 2016), a repeated measures model was used, with treatment, sampling year, and a treatment × sampling year interaction as the fixed effects. Blocks were included as random effects in all models. Assumptions were assessed for each model, and log transformations were made where necessary to meet assumptions. A significance level of 0.05 was used during all analyses.

RESULTS

Aboveground Biomass and Plant Species Composition

Fourteen yr after surface biosolids application, total aboveground biomass in the biosolids plots was nearly two times greater than in control plots (Table 1). An examination of biomass by

Table 1
Aboveground biomass (kg ha^{-1}) of forb and grass functional groups 14 yr after biosolids application on the study site in the Central Interior of British Columbia. Standard error of the mean is shown in brackets ($n = 4$).

Treatment	Aboveground biomass (kg ha^{-1})		
	Forbs	Grasses	Total
Biosolids	274 (147)	1267 (158)	1541 (133)
Control	235 (65)	556 (124)	791 (64)
P value	0.59	0.01	< 0.01

functional group showed that the grass biomass was more than double in the biosolids plots ($P = 0.01$) and forbs produced similar aboveground biomass ($P = 0.59$) in the biosolids and control treatments. However, the proportion of forbs in the total biomass was 30% in control and 18% in biosolids treatment. Furthermore, forbs were present in $100\% \pm 0.5\%$ and $87\% \pm 6\%$ of Daubenmire frames in the control and biosolids treatments, respectively. Frequency of common dandelion (*Taraxacum officinale* G.H. Weber ex Wiggers) and field milk-vetch (*Astragalus agrestis* Dougl. ex G. Don) was $51\% \pm 34\%$ and $31\% \pm 13\%$ in biosolids plots as compared with $3\% \pm 2\%$ and $69\% \pm 21\%$ in the control.

The cover of late-seral bluebunch wheatgrass was significantly greater in the biosolids plots 14 yr after a single biosolids application and protection from grazing, at 6% cover in the biosolids plots and < 0.1% in the control (Table 2). Since the beginning of the experiment, bluebunch wheatgrass cover was greatest in the biosolids treatment ($P < 0.001$) and there was a significant Year effect ($P = 0.003$) because cover in biosolids plots was the greatest in 2006 (Fig. 1a). The bluebunch wheatgrass plants growing in the biosolids plots in 2016 were greater in circumference and height and nearly five times greater in standing biomass than bluebunch wheatgrass plants in the control plots (Table 3). The number of flower stalks per bluebunch wheatgrass plant, an indicator of reproductive vigor, was not affected by biosolids application (see Table 3).

The most abundant grass species at the site was early-seral needle-and-thread grass (see Table 2; Fig. 1c). A significant Year × Treatment interaction ($P < 0.001$; Fig. 1b) was explained by a sharp increase in cover 3 yr after biosolids application (2005) while decreasing in the control. By 2016, needle-and-thread grass covered 33% and 31% of biosolids and control plots, respectively (see Table 2; Fig. 1c). Kentucky bluegrass was significantly greater in the biosolids treatment in 2016, covering 26% of biosolids plots and < 1% of control plots (see Table 2). Kentucky bluegrass was almost nonexistent on the study plots in 2002, yet in 2016, 76% of the increased perennial cover in the biosolids plots as compared with the control was attributed to Kentucky bluegrass, as indicated by the significant Year × Treatment interaction ($P < 0.001$; see Fig. 1b).

Besides bluebunch wheatgrass and Kentucky bluegrass, the only other plant species that were significantly affected by the treatment were pussytoes (*Antennaria* spp.), which were more abundant in the control plots (see Table 2). The cover of midseral bunch grass Nevada bluegrass (*Poa secunda* ssp. *juncifolia* Scribn.), late-seral Junegrass, early-seral Sandberg's bluegrass (*Poa secunda* ssp. *secunda* Presl), Prairie sagewort, field milk-vetch, nodding onion (*Allium cernuum* Roth), and common dandelion were not significantly affected by biosolids (see Table 2).

Soil Cover

In 2016, the biosolids treatment demonstrated significantly less exposed soil and less microbiotic crust, when compared with the control, but significantly greater cover of litter (Fig. 2). The sudden disappearance of microbiotic crust in the biosolids plots in 2003 (i.e., 1 yr after biosolids application) was due to it being buried by the applied biosolids material and explained the significant Year × Treatment interaction ($P < 0.001$; see Fig. 2b). In the following years, the biosolids material was in turn covered by litter and the microbiotic crust remained at negligible levels in the biosolids plots ($0.4\% \pm 0.2\%$ cover in biosolids and $31\% \pm 3\%$ in control; see Fig. 2b). Litter cover, despite being similar in both treatments for the first 3 yr (2002–2005), increased dramatically in the biosolids treatment in 2006 (Year × Treatment $P < 0.001$) and remained at significantly greater levels in 2016 relative to the control ($89\% \pm 3\%$ cover in biosolids and $35\% \pm 4\%$ in the control; see Fig. 2c). Bare soil decreased in both treatments in the first 3 yr (Year × Treatment $P =$

Table 2

Percent cover for individual plant species 14 yr after biosolids application on the study site in the Central Interior of British Columbia. Standard error of the mean are shown in brackets ($n = 4$).

Category	Species ¹	Origin	Late seral ²	Treatment		P value
				Biosolids	Control	
Perennial grasses	Needle-and-thread grass (<i>Hesperostipa comata</i>)	Native	< 1.0	33 (5.3)	31 (4.3)	0.32
	Junegrass (<i>Koeleria macrantha</i>)	Native	3.0–9.9	2 (1.0)	7 (1.5)	0.07
	Nevada bluegrass (<i>Poa secunda</i> ssp. <i>juncifolia</i>)	Native	—	7 (5.9)	0 (0.2)	0.16
	Kentucky bluegrass (<i>Poa pratensis</i>)	Agronomic	< 1.0%	26 (6.9)	0 (0.1)	< 0.01
	Sandberg's bluegrass (<i>Poa secunda</i> ssp. <i>secunda</i>)	Native	—	1 (0.8)	3 (1.0)	0.20
	Bluebunch wheatgrass (<i>Pseudoroegneria spicata</i>)	Native	> 25.0	6 (3.2)	0 (0.2)	0.04
Perennial forbs	Nodding onion (<i>Allium cernuum</i>)	Native	—	4 (3.3)	6 (4.7)	0.41
	Field milk-vetch (<i>Astragalus agrestis</i>)	Native	—	3 (1.0)	4 (1.4)	0.43
	Pussytoes (<i>Antennaria</i> spp.)	Native	3.0–9.9	0 (0.3)	8 (2.0)	0.02
	Prairie sagewort (<i>Artemisia frigida</i>)	Native	3.0–9.9	1 (0.5)	3 (0.9)	0.16
	Common dandelion (<i>Taraxacum officinale</i>)	Exotic	—	4 (2.4)	0 (0.0)	0.06

¹ Only species with > 2.5% cover in either biosolids or control plots are included in the table.

² Late seral column provides cover values (%) for certain native species expected to be found in a late seral plant community at this site according to Coupé and Iverson (2014).

0.008) and remained low in both treatments in 2016 ($1\% \pm 0.4\%$ cover in biosolids and $6\% \pm 2\%$ in the control; see Fig. 2d).

Forage Quality

Forage uptake of all macronutrients, N, S, Ca, K, Mg, and P (Fig. 3b); most micronutrients, B, Cu, Mn, Mo, and Zn; and Al (Fig. 4b) were significantly higher in plants from the biosolids amended plots. Only the uptake of Fe and Na was not different between treatments. In contrast to the nutrient uptake values, only the concentrations of P and Cu were significantly greater in the plants from the biosolids plots (see Fig. 3a), while N and Mg concentrations were greater in plants from the control plots (see Fig. 4a). Biosolids application was not found to have an effect on the Cu:Mo ratio of forage (Table 4). No other element showed a

difference in tissue concentration between biosolids and control plots. Concentrations of Al, Fe, and Na in plants on control plots, however, represented a 61%, 83%, and 126% respective increase over the concentrations in plants grown in biosolids plots. The plant biomass C:N ratio of 39.5 in the biosolids treatment (Table 5) was significantly higher than in the control (33.8) due to a higher N concentration in plants from control plots (see Fig. 3a).

Despite a 34% increase in total protein per ha (concentration $\times$ yield) in the biosolids plots relative to the control (see Table 4), the difference between treatments was not significant ($P = 0.05$). Protein concentrations, however, were significantly higher in the control plots. No long-term treatment effect of biosolids application on ADF levels was found. Similarly, no differences were found in cellulose or lignin (the components of ADF) content between biosolids and control treatments (see Table 4).

Aboveground and Thatch C Stocks

While biosolids application did not affect the soil C stock 14 yr after application (Avery et al., 2018), aboveground plant biomass (i.e., vegetation) C stock was nearly double in the biosolids plots as compared with the control plots (688 kg ha^{-1} and 358 kg ha^{-1} , respectively; see Table 5). In addition to mineral soil and aboveground biomass C, we characterized an additional C stock—the layer of thatch found beneath Kentucky bluegrass. The thatch covered around 25% of the biosolids plots to an average depth of 3.8 cm and was composed of 0.6–17% roots by volume (6% on average). It was not present on the control plots. The thatch on the biosolids plots represented a C stock of 6.4 Mg ha^{-1} (5.6 Mg ha^{-1} in the enmeshed soil held in the thatch and 0.8 Mg ha^{-1} in the roots forming the thatch layer).

DISCUSSION

Aboveground Biomass and Plant Species Composition

Fourteen yr after the biosolids application in our study (Jesmond site), there was substantially greater perennial grass cover in the

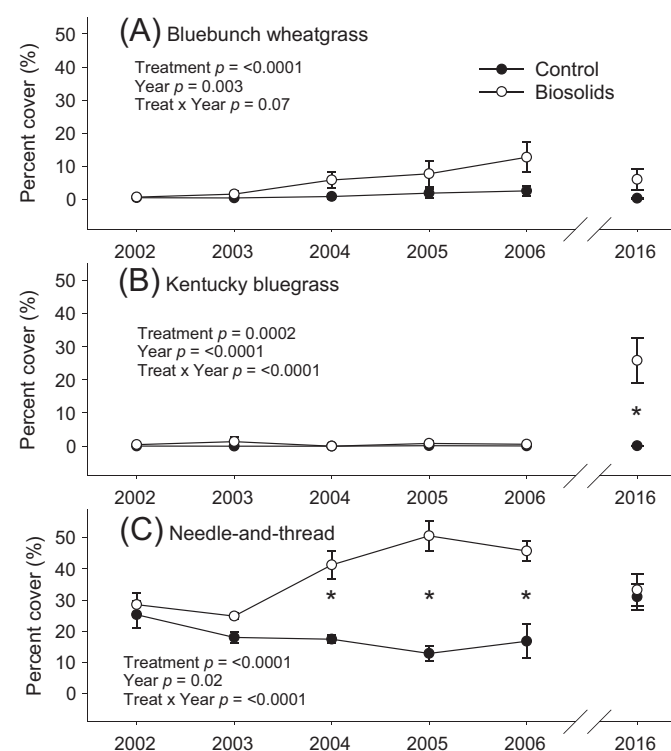


Figure 1. Cover of the three most dominant perennial grasses (bluebunch wheatgrass [A], Kentucky bluegrass [B], and needle-and-thread [C]) from 2002 to 2006 and another measurement in 2016 from biosolids and control plots. Error bars represent the standard error of the mean ($n = 4$). *Indicates a significant treatment effect at $P < 0.05$.

Table 3

Vigor indicators of bluebunch wheatgrass plants growing in biosolids and control treatments 14 yr after biosolids application on the study site in the Central Interior of British Columbia. Standard error of the mean are shown in brackets ($n = 4$).

Indicator of plant vigor	Biosolids	Control	P value
Circumference (cm)	37.7 (9.55)	22.1 (6.39)	0.03
Height (cm)	45.9 (4.12)	35.6 (3.06)	0.03
Number of flower stalks per plant	17.6 (6.45)	7.3 (3.20)	0.23
Standing biomass (g per plant)	39.9 (15.26)	8.0 (3.54)	0.08

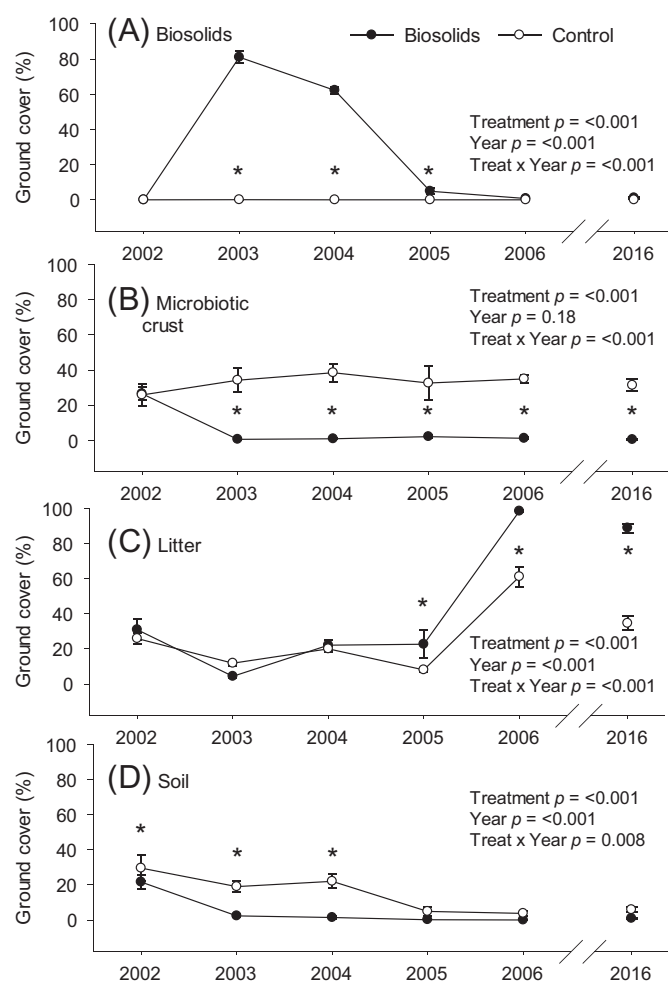


Figure 2. Percent of ground occupied by biosolids (A), bare soil (B), microbial crust (C), and litter (D) in biosolids and control plots from 2002 to 2006 and 2016. Error bars represent the standard error of the mean ($n = 4$). *Indicates a significant treatment effect at $P < 0.05$.

biosolids plots compared with the control plots (75% and 41% cover, respectively). In contrast, the total cover of native forbs was lower in the biosolids than control plots (13% and 24%, respectively). Effects of the biosolids application on plant cover were also noted in 2006 (4 yr post application), as reported by Newman et al. (2014), who attributed the increase in perennial grass cover to a combination of adequate levels of precipitation and absence of cheatgrass on site at the time of biosolids application. These two features distinguished the Jesmond site from two other sites included in the study by Newman et al. (2014), which were found in drier locations in the Interior of BC and also had high levels of cheatgrass cover at the time of application. At these two sites, biosolids led to an increase of exotic forbs and annual grasses at the expense of perennial grasses over 4 yr, suggesting that the results of this study may be limited to rangeland sites with similar levels of precipitation and plant communities at similar seral stages.

Although biosolids have been shown to increase forage production in the short term (Jurado and Wester, 2001; Wallace et al., 2009), fewer studies have found similarly long-lasting increases in forage productivity, especially as a result of a single biosolids application, as was observed in our study. Sullivan et al. (2006) found long-term increases in aboveground biomass with increasing application rates (2.5, 5, 10, 21, and 30 Mg ha⁻¹) on a semiarid shortgrass steppe rangeland dominated by native perennials western wheatgrass (*Pascopyrum smithii* [Rydb] A. Löve) and

needle-and-thread grass. The 21 Mg ha⁻¹ rate, which is closest to the rate in our study, achieved a 60% increase in aboveground biomass 12 yr after the application.

Another long-term (22-yr) study on the effects of organic waste amendments (composted manure, composted biosolids, and fresh manure) to a native mixed-grass prairie in Wyoming noted increases in native rangeland plant productivity 2 yr after applications of composted manure and composted biosolids (Blumenthal et al., 2017). However, 17 and 22 yr after the application, the differences in native plant productivity had given way to increases in cheatgrass (*Bromus tectorum* L.), which had significantly greater biomass in the composted manure treatment and followed the same trend, though not significantly, in the biosolids and fresh manure treatments. The authors highlighted that the full effects of organic waste application on rangeland plant species composition can manifest gradually and persist for decades, which highlights the importance of long-term studies when organic amendments are applied in semiarid environments.

In British Columbia, mesic to submesic grasslands in the IDFXm biogeoclimatic subzone, such as our study site, are expected to be dominated (> 25% cover) by bluebunch wheatgrass, accompanied

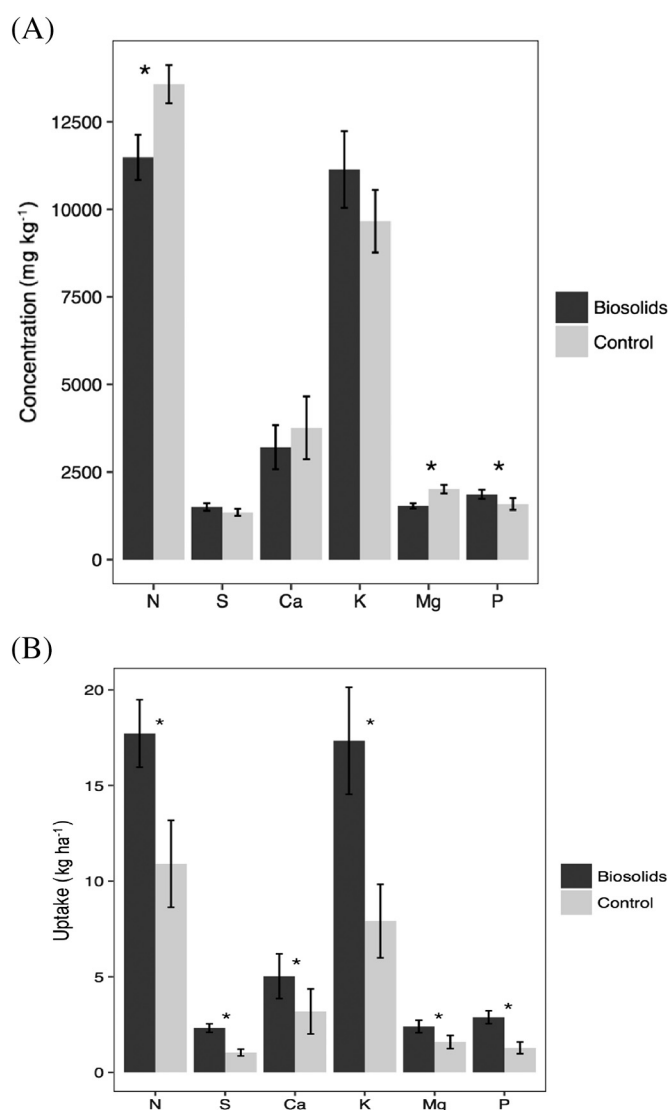


Figure 3. Concentration (A) and uptake (B) of plant macronutrients (N, S, Ca, Mg, K, and P) in plots with and without biosolids application 14 yr after application on the study site in the Central Interior of British Columbia. Error bars represent the standard error of the mean ($n = 4$). *Indicates a significant treatment effect at $P < 0.05$.

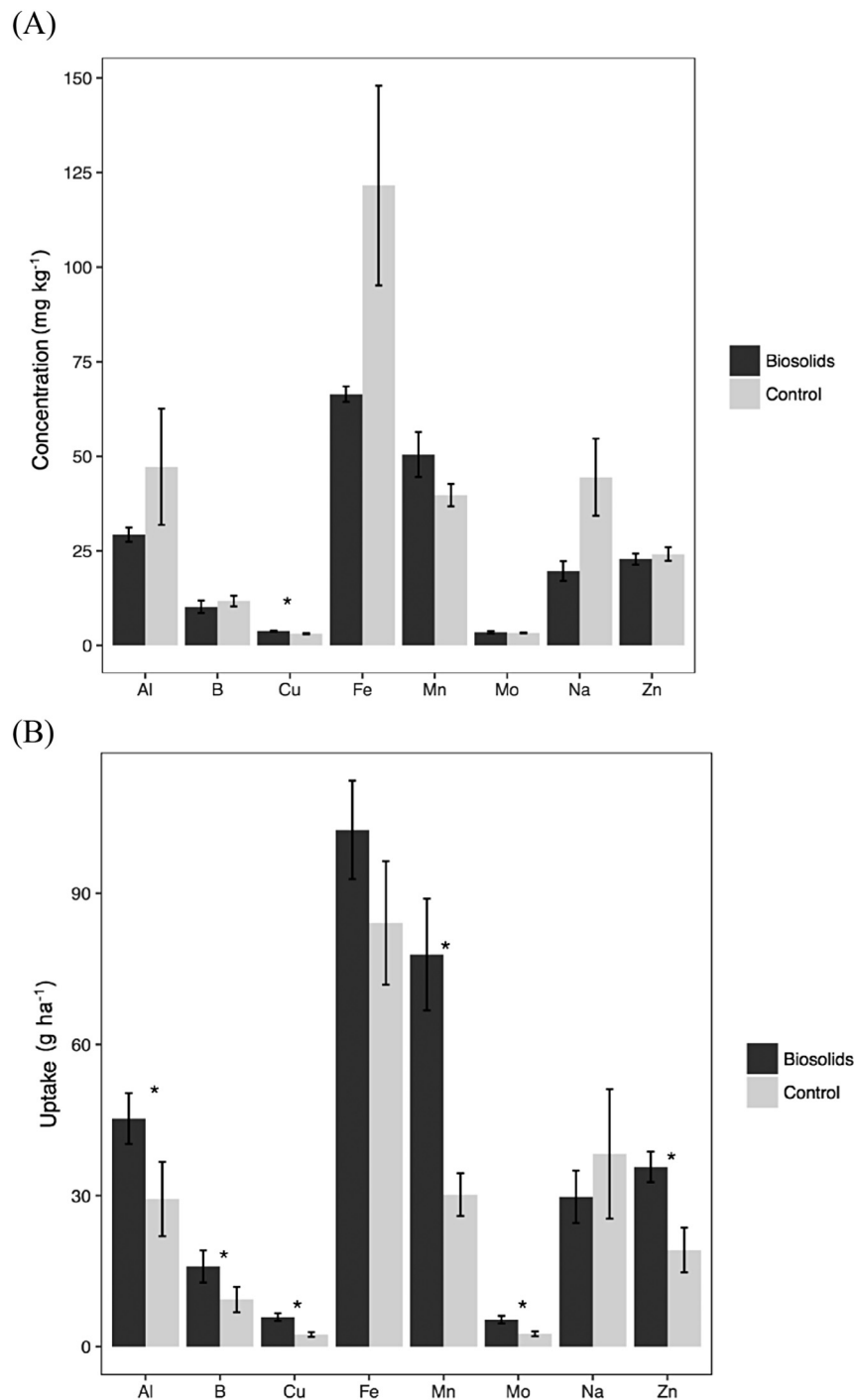


Figure 4. Concentration (A) and uptake (B) of plant micronutrients (B, Cu, Fe, Mn, Mo, Na, Zn) and Al in plots with and without biosolids application 14 yr after application on the study site in the Central Interior of British Columbia. Error bars represent the standard error of the mean ($n = 4$). *Indicates a significant treatment effect at $P < 0.05$.

Table 4

Forage quality in biosolids and control plots 14 yr after biosolids application on the study site in the Central Interior of British Columbia. Standard error of the mean is shown in brackets ($n = 4$).

Forage quality indicator	Biosolids	Control	P value
Protein (%)	7.0 (0.47)	10.2 (0.52)	0.01
Protein (kg ha ⁻¹)	109.1 (11.35)	81.3 (17.46)	0.05
Fiber (ADF ¹) (%)	38.9 (1.13)	38.7 (1.82)	0.86
Cellulose (%)	34.7 (1.22)	33.9 (1.68)	0.29
Lignin (%)	4.2 (0.23)	4.8 (0.15)	0.13
Cu:Mo	1.2 (0.11)	1.0 (0.06)	0.23

¹ ADF, acid detergent fiber.

by a diverse grass, forb, and cryptogamic community (Coupé and Iverson, 2014). Although the cover of bluebunch wheatgrass in the biosolids plots remained well below 25%, its increased cover in the biosolids plots has been evident 2 yr after the application of biosolids (see Fig. 1a). Individual bluebunch wheatgrass plants can take 8–10 yr to recover to full production and reproductive capacity after intense grazing and can take longer if additive stresses such as drought, insect herbivory, and competition are occurring (Mueggler, 1975; Anderson, 1991). Low-vigor plants will stop producing seed almost entirely. As such, time periods of a decade or longer and favorable environmental conditions are likely required

Table 5

Carbon concentration, content, and C:N ratio of soil, vegetation, and thatch C pools 14 yr after biosolids application in biosolids and control treatments. Standard error of the mean is shown in brackets ($n = 4$).

Carbon stock	Treatment		P value
	Biosolids	Control	
	mg kg ⁻¹		
Soil ¹	25.5 (2)	25.4 (2.5)	0.99
Vegetation	447 (1.9)	453 (0.8)	< 0.01
Thatch soil	92 (3.4)	n/a	n/a
Thatch roots	419 (7.9)	n/a	n/a
	Mg ha ⁻¹		
Soil ²	18.2 (0.1)	18.2 (0.1)	0.95
Vegetation	0.7 (0.1)	0.4 (0.1)	0.02
Thatch soil	5.6 (1.4)	n/a	n/a
Thatch roots	0.8 (0.2)	n/a	n/a
	C:N		
Soil ²	11 (0.0)	11 (0.0)	0.81
Vegetation	40 (2)	34 (1)	< 0.01
Thatch soil	12 (0.1)	n/a	n/a
Thatch roots	39 (5)	n/a	n/a

¹ Within 0- to 7.5-cm soil depth.

² Soil C content and C:N data are from Avery et al. (2018).

for bluebunch wheatgrass recovery (Anderson, 1991). The competition at our study site with needle-and-thread grass in both the biosolids and control plots, as well as Kentucky bluegrass in the biosolids plots, likely limited bluebunch wheatgrass recovery in our study.

In 2016, 76% of the increased perennial grass cover in the biosolids plots relative to the control was attributed to Kentucky bluegrass, which was at < 1% cover in 2002 (see Fig. 1b). Kentucky bluegrass is a cool-season perennial that spreads though the rhizomes have good forage value, which is not the point. It has garnered much attention due to its prolific expansion in the Northern Great Plains of the United States and the southern Canadian Prairies over the past 2–3 decades, as well as its ability to transform the structure and function of the ecosystem via its thatch (White et al., 2012; Toledo et al., 2014; DeKeyser et al., 2015; Bork et al., 2017; Otfinowski et al., 2017; Sanderson et al., 2017). The presence of Kentucky bluegrass on the biosolids plots in our study (and the relative absence of Kentucky bluegrass on the control) suggests that the biosolids application, in the context of 14 yr of rest from grazing, has encouraged the rise in cover of this plant species. The Shannon-Wiener diversity index was 25% higher in the control (1.9 ± 0.5 ; $P = 0.02$) as compared with biosolids plots (data not shown) when only native species were considered, which could have been attributed to the rise in Kentucky bluegrass cover. It is possible that Kentucky bluegrass benefited from increased available N and P in the biosolids plots, much like other cool season grasses (Vinton and Goergen, 2006; Blumenthal et al., 2008; Gioria and Osborne, 2014). Another possible explanation could be that biosolids applied to the soil surface reduced evaporation, which then allowed for more plant available water early in the growing season, a mechanism suggested by Blumenthal et al. (2008). Indeed, at our site, increased soil water was measured in May and June 2016 in the biosolids plots (Avery et al., 2018) and the accumulation of litter and reduced bare soil exposure (see Fig. 2) likely contributed to increased soil water content in the biosolids plots, which in turn may have supported the spread of Kentucky bluegrass. It should be noted that these conditions occurred in the absence of grazing, which allowed for the accumulation of litter. It is possible that Kentucky bluegrass would not have thrived in the biosolids plots had the regular removal of litter via grazing been occurring.

Forage Quality

Despite showing little difference in concentration values between the biosolids forage and control, most nutrients had greater

uptake values (nutrient concentration $\times$ yield) in the biosolids plots due to the more than double aboveground biomass on the biosolids plots. Phosphorus was the only macronutrient that had significantly greater concentration in the plants growing in biosolids plots. P was also the only macronutrient that was significantly more available in biosolids amended soil (Avery et al., 2018). The fact that P is more concentrated in plants that received biosolids 14 yr ago may suggest that the plants continue to benefit from elevated levels of soil P, which is typically a limiting nutrient in semiarid rangeland ecosystems. Copper, the only other nutrient more concentrated in plant tissues from biosolids plots, was more available in the biosolids amended soil at two out of four sampling dates (Avery et al., 2018).

In contrast, the forage on the control plots had higher tissue concentrations of N and Mg relative to the biosolids amended forage, while no difference was found in plant-available levels of N and Mg between biosolids amended soil and control soil (Avery et al., 2018). A possible explanation for this is the “dilution effect,” a well-known effect reviewed by Jarrell and Beverly (1981), wherein increases in yield result in the decrease in concentration of nutrients in the plant tissue, where available soil nutrient levels are held constant. It is possible that greater soil moisture and/or soil P levels on the biosolids amended plots (Avery et al., 2018) promoted the greater aboveground biomass on the biosolids plots but also caused a dilution effect on N and Mg levels.

The higher protein concentration in the forage from the control plots relative to the forage from the biosolids plots (see Table 2) was in accordance with the finding of higher N concentration discussed earlier (see Fig. 3a). It can be explained in the same way as the higher N concentration: likely the result of a dilution effect of the twofold increase in aboveground biomass in the biosolids plots. An average protein concentration of 12% is common for bunchgrass forage in southern and central BC (McDougall et al., 2008), and average protein content ranges from 70 to 156 kg ha⁻¹, which suggests that both control and biosolids plots in our study were below regional average protein concentrations but close to the average total contents of protein per hectare.

A second possible explanation for the higher N, Mg, and protein concentrations in the control plots is the increased proportion of forbs. Forbs represented 18% of the biomass from the biosolids plots and 30% from the control plots (see Table 1). Forbs have been shown to have higher N and lower C:N ratios than grasses (Wilsey and Wayne Polley, 2006). It has also been shown that within plant functional groups, different species on the same site often have different N concentrations due to different uptake mechanisms or differences in rooting depth (Koerselman and Meuleman, 1996). These niche characteristics allow for coexistence of a diversity of species, especially on resource-limited sites (Harpole and Tilman, 2007). The increase in forbs could also account for the high variability for certain elements (Al, Fe, and Na) in the control plots.

Although other studies (McFarland et al., 2007) have shown that biosolids can improve forage quality by increasing protein content and digestibility in the short term, the effect was not evident in our study conducted 14 yr after biosolids application. Grass and legume forages with ADF values under 35% are considered good quality (Van Saun, 2013). The ADF levels found in the forage at our site (see Table 2) were therefore marginally higher than ideal in both the biosolids treatment and the control plots. It has been reported that Cu:Mo below 2 can result in Cu deficiency (Miltimore and Mason, 1971; Gardner et al., 2012). Biosolids application did not have a long-term effect on the forage Cu:Mo ratio, which was 1.2 in the biosolids plots and 1.0 in the control 14 yr after the application (see Table 2). In a study by Pierce et al. (1998) on semiarid shrubland, biosolids increased the Cu:Mo ratio from below 1.2 to above 2 for western wheatgrass and bluebunch wheatgrass 1 yr after application at low rates (10 or 15 Mg ha⁻¹) and 2 yr after application at

intermediate rates (25–40 Mg ha⁻¹). It is possible that any positive effects of biosolids application on forage Cu:Mo ratio are relatively short term.

Soil Cover

The decrease in bare soil in both treatments in the first 3 yr of the experiment was likely a result of the exclusion of grazing, which allowed the plant community to recover. The greater litter cover on the biosolids plots, however, contributed to the significantly lower level of bare soil on the biosolids plots as compared with the control plots in 2016. Reducing bare soil is important to proper ecosystem function in semiarid rangelands because it leads to a lower susceptibility of erosion and evaporation. Microbiotic crust plays an important role in semiarid bunchgrass ecosystems by protecting the soil in the interspaces between bunchgrasses in ecosystems that are typically too dry, or degraded, for significant accumulation of plant litter (Wikeem and Wikeem, 2004). In the biosolids plots, however, the nearly continuous cover of plant litter (89%) and high levels of aboveground perennial grass biomass (see Fig. 2; Table 1) provided that ecosystem service.

Aboveground and Thatch C Stocks

The thatch of Kentucky bluegrass represented a substantial C stock: the matted thatch rhizomes contributed an additional 775 kg of C per ha, which is even greater than the C stored in the aboveground biomass on biosolids plots (688 kg ha⁻¹). Studies rarely report thatch C stocks, and when the belowground C stock is reported, it mainly focuses on roots. For example, a study by Zhang et al. (2018) carried out on a rough fescue (*Festuca campestris* Rydb.) grassland with codominant Parry's oat grass (*Danthonia parryi* Scribn.) and Kentucky bluegrass reported root C stocks (from the 0- to 15-cm soil depth) that were three times greater than the standing biomass C stock.

Kentucky bluegrass commonly creates a thatch layer in grassland ecosystems, likely due to the combination of its vigorous aboveground production and the relatively slow belowground decomposition that occurs in moisture-limited grasslands (Murray and Juska, 1970; Smiley, 1981). In our study, the absence of grazing for 14 yr likely has contributed to the accumulation of thatch at the soil surface. Although the biosolids application did not result in any changes in mineral soil C stocks (Avery et al., 2018), there was certainly an accumulation of aboveground biomass (i.e., vegetation) C and thatch C on biosolids plots (see Table 5). In addition, the soil enmeshed within the thatch had a higher C concentration (9.2%) as compared with the soil below the thatch (2.6%; see Table 5), likely due to the high concentration of roots and partially decomposed organic matter in close contact with mineral soil particles in the thatch. Although the accumulation of thatch C is partly an outcome of grazing exclusion, the application of biosolids is mainly done to improve forage production. It is also done in a restoration context where grazing is excluded for long periods of time. Therefore, the long-term results of increased thatch and cover of Kentucky bluegrass with an absence of grazing are important for many scenarios where biosolids are applied in semiarid environments.

IMPLICATIONS

Assessments of plant species composition, forage quantity and quality, and C stocks 14 yr after a single surface biosolids application at 20 Mg ha⁻¹ provided a unique opportunity to quantify long-term effects of biosolids application to semiarid grassland. The existing literature on organic matter amendments to rangeland is dominated by short-term studies, despite indications that the effects of these practices may continue to unfold over decades

(Blumenthal et al., 2017). The long-term changes noted in this study in plant community composition, notably the continued persistence of bluebunch wheatgrass and recent invasion of Kentucky bluegrass, highlight the importance of remeasuring long-term experiments. In addition, the long-term effects of biosolids application to rangeland appear to be highly influenced by characteristics of the site (i.e., moisture regime, plant community at time of application) (Newman et al., 2014; Blumenthal et al., 2017). As such, more long-term studies are necessary in order to better understand the relationships between site characteristics and the effects of biosolids application, which would ultimately allow land managers to better predict the long-term effects of applications to their specific rangeland site.

The results of this study may be interpreted in several ways, depending on priorities of different stakeholder groups and management objectives for the rangeland site. For example, the documented long-term increase in forage production, reduction in bare soil, and resulting increase in soil moisture would be of value to ranchers or stakeholders valuing forage production. While the decrease of protein concentration in biosolids forage may be a concern to this group, the greater plant uptake of most nutrients in the biosolids plots (including N) due to greater aboveground biomass would be an additional benefit. The finding of greater bluebunch wheatgrass cover in the biosolids plots would likely be prioritized as a positive outcome by stakeholders focused on rangeland restoration and native species conservation, while the increase of Kentucky bluegrass cover may be of concern, given its potential ability to alter ecosystem functioning. Kentucky bluegrass, however, is generally considered a high-quality forage species, so the interpretation of this result will, again, depend on values held by the stakeholder. As such, this work suggests that when considering the application of biosolids or other organic matter amendments to rangeland, it is crucial to understand the management goals of the site, along with the likely interactions between unique site characteristics and biosolids, in order to assess whether the outcomes of the application will support the site's management goals in the long-term.

ACKNOWLEDGMENTS

We thank Lawrence Joiner, local rancher, for providing access to his fields, and students Michael Oh, Karly Vanichuk, Thea Rodgers, and Ivan Nesic, as well as Roz Kempe (Metro Vancouver), for their help in the field and laboratory. We also acknowledge Metro Vancouver and BC Ministry of Forests for providing funding for this study in 2016.

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